Arjuna JEE 2027

JEE MAINS & IIT JEE ADVANCED

MATHEMATICS

SEQUENCE & SERIES



DIBY (DO IT BY YOURSELF)

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DIBY-01

Let a_n be the nth term of an $A.P.$. If $S_n = a_1 + a_2 + a_3 + \cdots . + a_n = 700, a_6 = 7$ and $S_7 = 7$, then a_n is equal to: **[JEE Mains-2025]**

(A) 56 (B) 65 (C) 64 (D) 70

[Ans. : (C)]

DIBY-02

If the first term of an A.P. is 3 and the sum of its first four terms is equal to one-fifth of the sum of the next four terms, then the sum of the first 20 terms is equal to **[JEE Mains-2025]**

(A) –120 (B) –1200 (C) –1020 (D) –1080

[Ans. : (D)]

DIBY-03

The roots of the quadratic $3x^2 - px + q = 0$ are 10th and 11th terms of an arithmetic progression with common difference $\frac{3}{2}$. If the sum of the first 11 terms of this arithmetic progression is 88, then $q - 2p$ is equal to **[JEE Mains-2025]**

[Ans. : 474]

DIBY-04

In an arithmetic progression, if $S_{40} = 1030$ and $S_{12} = 57$, then $S_{30} - S_{10}$ is equal to

[JEE Mains-2025]

(A) 505 (B) 525 (C) 510 (D) 515

[Ans. : (D)]

DIBY-05

If $a_1, a_2, a_3, \ldots$ are in A.P. such that $a_1 + a_7 + a_{16} = 40$, then the sum of the first 15 terms of this A.P. is

[JEE Mains-2019]

(A) 200 (B) 280 (C) 120 (D) 260

[Ans. : (A)]

DIBY-06

If 19th terms of non-zero A.P. is zero, then its (49th term) : (29th term) is:

[JEE Mains-2019]

(A) 4 : 1 (B) 1 : 3 (C) 3 : 1 (D) 2 : 1

[Ans. : (C)]

DIBY-07

Let S_n be the sum of the first n terms of an arithmetic progression. If $S_{3n} = 3S_{2n}$, then the value of $\frac{S_{4n}}{S_{2n}}$ is: **[JEE Mains-2021]**

(A) 2 (B) 6 (C) 8 (D) 4

[Ans. : (B)]

DIBY-08

Let $a_1, a_2, a_3, \ldots$ be an A.P. If $\frac{a_1+a_2+\cdots+a_{10}}{a_1+a_2+\cdots+a_p} = \frac{100}{p^2}, p \neq 10$, then $\frac{a_{11}}{a_{10}}$ is equal to:

[JEE Mains-2021]

(A) $\frac{21}{19}$ (B) $\frac{100}{121}$ (C) $\frac{121}{100}$ (D) $\frac{19}{21}$

[Ans. : (A)]

DIBY-09

If $\log_3 2, \log_3(2^x - 5), \log_3\left(2^x - \frac{7}{2}\right)$ are in an arithmetic progression, then the value of x is equal to

[JEE Mains-2021]

[Ans. : 3]

DIBY-10

If the 10th term of an A.P. is $\frac{1}{20}$ and its 20th term is $\frac{1}{10}$, then the sum of its first 200 terms is

[JEE Mains-2020]

(A) 100 (B) $50\frac{1}{4}$ (C) $100\frac{1}{2}$ (D) 50

[Ans. : (C)]

DIBY-11

If the sum of the series $20 + 19\frac{3}{5} + 19\frac{1}{5} + 18\frac{4}{5} + \cdots$ upto n^{th} term is 488 and the nth term is negative, then **[JEE Mains-2020]**

(A) $n = 41$

(B) n^{th} **term is** $-4\frac{2}{5}$

(C) $n = 60$

(D) n^{th} **term is -4**

[Ans. : (D)]

DIBY-12

Five numbers are in A.P., whose sum is 25 and product is 2520. If one of these five numbers is $\frac{-1}{2}$, then the greatest number amongst them is **[JEE Mains-2020]**

(a) $\frac{21}{2}$ **(B)** 16 **(C)** 27 **(D)** 7

[Ans. : (B)]

DIBY-13

If $3^{2\sin 2\alpha-1}, 14$ and $3^{4-2\sin 2\alpha}$ are the first three terms of an A.P. for some α, then the sixth term of this A.P. is: **[JEE Mains-2020]**

(A) 65 (B) 78 (C) 81 (D) 66

[Ans. : (D)]

DIBY-14

If $a_1, a_2, a_3.$ and $b_1, b_2, b_3.$ are in A.P., and $a_1 = 2, a_{10} = 3$, $a_1 b_1 = 1 = a_{10} b_{10}$, then $a_4 b_4$ is equal to. **[JEE Mains-2022]**

(A) $\frac{35}{27}$ (B) 1 (C) $\frac{27}{28}$ (D) $\frac{28}{27}$

[Ans. : (D)]

DIBY-15

A software company sets up m number of computer systems to finish an assignment in 17 days. If 4 computer systems crashed on the start of the second day, 4 more computer systems crashed on the start of the third day and so on, then it took 8 more days to finish the assignment. The value of m is equal to: **[JEE Mains-2024]**

(A) 125 (B) 150 (C) 180 (D) 160

[Ans. : (B)]

DIBY-16

Consider two sets A and B, each containing three numbers in $A.P$. Let the sum and the product of the elements of A be 36 and p respectively and the sum and the product of the elements of B be 36 and q respectively. Let d and D be the common differences of AP's in A and B respectively such that $D = d + 3, d > 0$. If $\frac{p+q}{p-q} = \frac{19}{5}$, then $p - q$ is equal to [JEE Mains-2025]

(A) 600 (B) 450 (C) 630 (D) 540

[Ans. : (D)]

DIBY-17

Consider an A. P. of positive integers, whose sum of the first three terms is 54 and the sum of the first twenty terms lies between 1600 and 1800. Then its 11th term is: [JEE Mains-2025]

(A) 84 (B) 90 (C) 108 (D) 122

[Ans. : (B)]

DIBY-18

Number of 4 digit integers less than 2800 which are either divisible by 3 or by 11 is equal to ? **[JEE Mains-2023]**

[Ans. : 710]

DIBY-19

If where n is an even integer, is an arithmetic progression with common difference 1, and $\sum_{i=1}^{n} a_i = 192, \sum_{i=1}^{n/2} a_{2i} = 120,$, then n is equal to **[JEE Mains-2022]**

(A) 48 (B) 96 (C) 92 (D) 104

[Ans. : (B)]

DIBY-20

Let S_1 be the sum of first $2n$ term of an arithmetic progression. Let S_2 be the sum of first $4n$ terms of the same arithmetic progression. If $(S_2 - S_1)$ is 1000, then the sum of the first $6n$ terms of the arithmetic progression is equal to

[JEE Mains-2021]

(A) 5000 (B) 1000 (C) 7000 (D) 3000

[Ans. : (D)]

DIBY-21

The sum of all natural numbers 'n' such that $100 < n < 200$ and H.C.F. $(91, n) > 1$ is

[JEE Mains-2019]

(A) 3221 (B) 3121 (C) 3203 (D) 3303

[Ans. : (B)]

DIBY-22

The series of positive multiples of 3 is divided into sets : {3}, {6, 9, 12}, {15, 18, 21, 24, 27}, ... Then the sum of the elements in the 11th set is equal to

[JEE Mains-2022]

[Ans. : 6993]

DIBY-23

Let $a_1, a_2, \ldots a_{30}$ be an A.P., $S = \sum_{i=1}^{30} a_i$ and $T = \sum_{i=1}^{15} a_{2i-1}$. If $a_5 = 27$ and $S - 2T = 75$, then a_{10} is equal to

[JEE Mains-2019]

(A) 52 (B) 57 (C) 47 (D) 42

[Ans. : (A)]

DIBY-24

Let $a_1, a_2, \ldots a_{2024}$ be an Arithmetic Progression such that $a_1 + (a_5 + a_{10} \quad + a_{15} + \cdots + a_{2020}) + a_{2024} = 2233$. Then $a_1 + a_2 + a_3 + \cdots + a_{2024}$ is equal to

[JEE Mains-2025]

[Ans. : 11132]

DIBY-25

If p^{th}, q^{th} and r^{th} term of an A.P. are respectively a, b and c, then the value of $a(q-r) + b(r-p) + c(p-q)$ is

(A) 0 (B) 1 (C) $p+q+r$ (D) abc

[Ans. : (A)]

DIBY-26

Find the value of the sum $S = 100^2 - 99^2 + 98^2 - 97^2 + 96^2 - 95^2 + \ldots + 2^2 - 1^2$

(A) 4950 (B) 10100 (C) 5150 (D) 5050

[Ans. : (D)]

DIBY-27

The sum of first 3 terms of an A.P. is 27 and the sum of their squares is 293, then first term and common difference of A.P. is

(A) First term $=$ 3, Common difference $= \pm 6$

(B) First term = 4, Common difference = 5

(C) First term = 4 or 14, Common difference = ± 5

(D) First term = 2 or 16, Common difference $= \pm 7$

[Ans. : (C)]

DIBY-28

The number of terms in an A.P. is even, the sum of the odd numbered terms is 24 & sum of the even numbered terms is 30 and the last term exceeds the first term by $10\frac{1}{2}$. Find the number of terms.

(A) 4 (B) 6 (C) 8 (D) 12

[Ans. : (C)]

DIBY-29

How many terms of the series, $20 + 19\frac{1}{3} + 18\frac{2}{3} + \cdots\ldots$ must be taken so that sum is 300. Explain the reason of double answer.

(A) $n = 25$ **or** $n = 36$**; the sum of the terms from** 26th **to** 36th **is zero.**

(B) $n = 20$ **or** $n = 30$**; the common difference becomes positive after the** 20th **term.**

(C) $n = 15$ **or** $n = 40$**; the terms after the** 15th **term are perfectly symmetrical.**

(D) $n = 25$ **only; the negative value of** $n = -36$ **is rejected.**

[Ans. : (A)]

DIBY-30

If the sum of the roots of the equation $ax^2 + bx + c = 0$ **is equal to the sum of the squares of their reciprocals, then show that** bc^2, ca^2, ab^2 **are in A.P.**

DIBY-31

A person is to count 4500 currency notes. Let a_n denote the number of notes he counts in the nth minute. If $a_1 = a_2 = \cdots = a_{10} = 150$ and $a_{10}, a_{11}, \ldots$ are in an AP with common difference -2, then the time taken by him to count all notes is

[AIEEE-2010]

(A) 24 minutes (B) 34 minutes (C) 125 minutes (D) 135 minutes

[Ans. : (B)]

DIBY-32

Let the sum of the first n terms of a non-constant A.P., $a_1, a_2, a_3 \ldots$ be $50n + \frac{n(n-7)}{2}A$, where A is constant. If d is the common difference of this A.P., then the ordered pair (d, a_{50}) is equal to:

[JEE Mains-2019]

(A) $(A, 50 + 45A)$

(B) $(50, 50 + 46A)$

(C) $(A, 50 + 46A)$

(D) $(50, 50 + 45A)$

[Ans. : (C)]

DIBY-33

Along a road lies an odd number of stones placed at intervals of 10 m. These stones have to be assembled around the middle stone. A person can carry only one stone at a time. A man carried out the job starting with the stone in the middle, carrying stones in succession, thereby covering a distance of 4.8 km. Then the number of stones is

(A) 15 (B) 29 (C) 31 (D) 35

[Ans. : (C)]

GEOMETRIC PROGRESSION

DIBY-34

Let $a_1, a_2, a_3, \ldots$ be a G. P. of increasing positive numbers. If $a_3 a_5 = 729$ and $a_2 + a_4 = \frac{111}{4}$, then $24(a_1 + a_2 + a_3)$ is equal to [JEE Mains-2025]

(A) 131 (B) 130 (C) 129 (D) 128

[Ans. : (C)]

DIBY-35

Let $a_1, a_2, a_3, \ldots$ be a G.P. of increasing positive terms. If $a_1 a_5 = 28$ and $a_2 + a_4 = 29$, then a_6 is equal to: [JEE Mains-2025]

(A) 812 (B) 526 (C) 784 (D) 628

[Ans. : (C)]

DIBY-36

Let a and b be two distinct positive real numbers. Let 11th term of a GP, whose first term is a and third term is b, is equal to p^{th} term of another GP, whose first term is a and fifth term is b. Then p is equal to [JEE Mains-2024]

(A) 20 (B) 25 (C) 21 (D) 24

[Ans. : (C)]

GEOMETRIC PROGRESSION

DIBY-37

Let $a_1, a_2, a_3, \ldots$ be a G.P. of increasing positive numbers. Let the sum of its 6th and 8th terms be 2 and the product of its 3rd and 5th terms be $\frac{1}{9}$.

Then $6(a_2 + a_4)(a_4 + a_6)$ is equal to [JEE Mains-2023]

(A) $2\sqrt{2}$ (B) 2 (C) $3\sqrt{3}$ (D) 3

[Ans. : (D)]

DIBY-38

In an increasing geometric progression of positive terms, the sum of the second and sixth terms is $\frac{70}{3}$ and the product of the third and fifth terms is 49. Then the sum of the 4th, 6th and 8th terms is: [JEE Mains-2024]

(A) 96 (B) 78 (C) 91 (D) 84

[Ans. : (C)]

GEOMETRIC PROGRESSION

DIBY-39

Find the four successive terms of a. G.P. of which the 2nd term is smaller than the first by 35 and the 3rd term is larger than the 4th by 560.

(A) $7, -28, 112, -448$

(B) $5, -20, 80, -320$

(C) $7, 28, 112, 448$

(D) $-\frac{35}{3}, -\frac{140}{3}, -\frac{560}{3}, -\frac{2240}{3}$

[Ans. : A, D]

DIBY-40

If a_1, a_2, a_3 are in G.P. such that $a_1 + a_2 + a_3 = 13$ and $a_1^2 + a_2^2 + a_3^2 = 91$. Find a and r.

(A) $(a = 1, r = 2)$ **or** $\left(a = 4, r = \frac{1}{2}\right)$

(B) $(a = 3, r = 1)$ **or** $\left(a = \frac{1}{3}, r = 9\right)$

(C) $(a = 2, r = 3)$ **or** $\left(a = 9, r = \frac{1}{2}\right)$

(D) $(a = 1, r = 3)$ **or** $\left(a = 9, r = \frac{1}{3}\right)$

[Ans. : (D)]

DIBY-41

If $S_1, S_2, S_3, \ldots . S_p$ are the sums of infinite G.P. whose first terms are $1, 2, 3, \ldots \ldots p$, and whose common ratios are $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots \ldots \frac{1}{p+1}$ respectively, then find the value of $S_1 + S_2 + S_3 + \cdots S_p$.

(A) $\frac{p(p+1)}{2}$ **(B)** $\frac{(p+1)(p+3)}{2}$ **(C)** $\frac{p}{2}(p+3)$ **(D)** $p(p+3)$

[Ans. : (C)]

DIBY-42

The sum of an infinite number of terms of a G.P. is 15 and the sum of their squares is 45. Find the series.

(A) $5, \frac{10}{3}, \frac{20}{9}, \frac{40}{27}, \ldots$ **(B)** $3, 2, \frac{4}{3}, \frac{8}{9}, \ldots$

(C) $5, \frac{5}{3}, \frac{5}{9}, \frac{5}{27}, \ldots$ **(D)** $3, \frac{6}{5}, \frac{12}{25}, \ldots$

[Ans. : (A)]

GEOMETRIC PROGRESSION

DIBY-43

If the sum of an infinite GP a, ar, ar^2, ar^3....., is 15 and the sum of squares of of its each term is 150, then the sum of of ar^2, ar^4, ar^6.....upto infinity, is:

[JEE Mains-2021]

(A) $\frac{1}{2}$ **(B)** $\frac{25}{2}$ **(C)** $\frac{5}{2}$ **(D)** $\frac{9}{2}$

[Ans. : (A)]

DIBY-44

Three numbers are in an increasing geometric progression with common ratio r. If the middle number is doubled, then the new numbers are in an arithmetic progression with common difference d. If the fourth term of GP is $3r^2$, then $r^2 - d$ is equal to:

[JEE Mains-2021]

(A) $7 + \sqrt{3}$ **(B)** $7 - \sqrt{3}$ **(C)** $7 + 3\sqrt{3}$ **(D)** $7 - 7\sqrt{3}$

[Ans. : (A)]

DIBY-45

If the sum of the second, third and fourth terms of a positive term. G.P. is 3 and the sum of its sixth, seventh and eighth terms is 243, then the sum of the first 50 terms of this G.P. is: **[JEE Mains-2020]**

(A) $\frac{2}{13}(3^{50}-1)$ **(B)** $\frac{1}{13}(3^{50}-1)$

(C) $\frac{1}{26}(3^{49}-1)$ **(D)** $\frac{1}{26}(3^{50}-1)$ **[Ans. : (D)]**

DIBY-46

The greatest positive integer k, for which 49^k+1 is a factor of the sum $49^{125}+49^{124}+\cdots+49^2+49+1$, is **[JEE Mains-2020]**

(A) 32 **(B)** 63 **(C)** 65 **(D)** 60

[Ans. : (B)]

DIBY-47

If the sum of the second, fourth and sixth terms of a G.P. of positive terms is 21 and the sum of its eighth, tenth and twelfth terms is 15309, then the sum of its first nine terms is: **[JEE Mains-2025]**

(A) 760 **(B)** 755 **(C)** 750 **(D)** 757

[Ans. : (D)]

DIBY-48

If three distinct numbers a, b, c are in G.P. and the equations $ax^2 + 2bx + c = 0$ and $dx^2 + 2ex + f = 0$ have a common root, then which one of the following statements is correct? [JEE Mains-2019]

(A) d, e, f are in A.P.

(B) $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in G.P.

(C) $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in A.P.

(D) d, e, f are in G.P.

[Ans. : (C)]

DIBY-49

Find the sum of the following series

(i) 5 + 55 + 555 + ... to n terms

(ii) 0.3 + 0.33 + 0.333 + ... to n terms

[Ans. : (i) $\frac{5}{9}\left[10\left(\frac{10^n-1}{9}\right) - n\right]$ (ii) $\frac{1}{3}\left[n - \frac{1}{9}\left(1 - \frac{1}{10^n}\right)\right]$]

DIBY-50

If $x = 1 + a + a^2 + a^3 + \ldots\ldots \infty$ and $y = 1 + b + b^2 + b^3 + \ldots\ldots \infty$ show that

$1 + ab + a^2b^2 + a^3b^3 + \ldots\ldots \infty = \frac{xy}{x+y-1}$ where $0 < a < 1$ and $0 < b < 1$.

DIBY-51

For the two positive number a, b if a, b and $\frac{1}{18}$ are in a geometric progression, while $\frac{1}{a}$, 10 and $\frac{1}{b}$ are in an arithmetic progression, then $16a + 12b$ is equal to

[JEE Mains-2023]

[Ans. : 3]

DIBY-52

If the first 3 consecutive terms of a geometrical progression are the roots of the equation $2x^3 - 19x^2 + 57x - 54 = 0$ find the sum to infinite number of terms of G.P.

[Ans. : $\frac{27}{2}$]

DIBY-53

Let x_1, x_2, x_3, x_4 be in a geometric progression. If $2, 7, 9, 5$ are subtracted respectively from x_1, x_2, x_3, x_4 then the resulting numbers are in an arithmetic progression. Then the value of $\frac{1}{24}(x_1 x_2 x_3 x_4)$ is:

[JEE Mains-2025]

(A) 72 (B) 18 (C) 36 (D) 216

[Ans. : 216]

DIBY-54

If p arithmetic mean are inserted between 5 and 41 so that ratio $\frac{A_3}{A_{p-1}} = \frac{2}{5}$, then find the value of p.

[Ans. : 11]

DIBY-55

If $a_1 a_2 a_3 \ldots\ldots a_n$ are n arithmetic means inserted between 7 and 2015 whose sum is 56616, then

(A) n is 56
(B) n is 28
(C) $a_{19} = 2029/3$
(D) $a_{19} = 36543/57$

[Ans. : (A,C)]

DIBY-56

The sides of a right angled triangle are in arithmetic progression. If the triangle has area 24, then what is the length of its smallest side?

[Ans. : 6]

DIBY-57

Let $S = 109 + \frac{108}{5} + \frac{107}{5^2} + \cdots + \frac{2}{5^{107}} + \frac{1}{5^{108}}$. Then the value of $\left(16S - (25)^{-54}\right)$ is equal to

[JEE Mains-2023]

[Ans. : 2175]

DIBY-58

Suppose $a_1, a_2, 2, a_3, a_4$ be in an arithmetico-geometric progression. If the common ratio of the corresponding geometric progression is 2 and the sum of all 5 terms of the arithmetico-geometric progression is $\frac{49}{2}$, then a_4 is equal to _____ .

[JEE Mains-2023]

[Ans. : 16]

DIBY-59

If $7 = 5 + \frac{1}{7}(5 + \alpha) + \frac{1}{7^2}(5 + 2\alpha) + \frac{1}{7^3}(5 + 3\alpha) + \cdots \infty$, then the value of α is :

[JEE Mains-2025]

(A) $\frac{6}{7}$ (B) $\frac{1}{7}$ (C) 6 (D) 1

[Ans. : (C)]

DIBY-60

Let $S = 2 + \frac{6}{7} + \frac{12}{7^2} + \frac{20}{7^3} + \frac{30}{7^4} + \cdots$. Then $4S$ is equal to [JEE Mains-2022]

(A) $\left(\frac{7}{3}\right)^2$ (B) $\frac{7^3}{3^2}$ (C) $\left(\frac{7}{3}\right)^3$ (D) $\frac{7^2}{3^3}$

[Ans. : (C)]

DIBY-61

If $\frac{6}{3^{12}} + \frac{10}{3^{11}} + \frac{20}{3^{10}} + \frac{40}{3^9} + \cdots + \frac{10240}{3} = 2^n \cdot m$, where m is odd, then mn is equal to [JEE Mains-2022]

[Ans. : 12]

DIBY-62

If $S = \frac{7}{5} + \frac{9}{5^2} + \frac{13}{5^3} + \frac{19}{5^4} + \cdots$, then $160S$ is equal to [JEE Mains-2021]

[Ans. : 305]

DIBY-63

If $S(x) = (1+x) + 2(1+x)^2 + 3(1+x)^3 + \cdots + 60(1+x)^{60}, x \neq 0$, and $(60)^2 S(60) = a(b)^b + b$, where $a, b \in N$, then $(a+b)$ equal to [JEE Mains-2024]

[Ans. : 3660]

DIBY-64

Let a and b be positive real numbers. If a, A_1, A_2, b are in arithmetic progression, a, G_1, G_2, b are in geometric progression and a, H_1, H_2, b are in harmonic progression, show that $\frac{G_1G_2}{H_1H_2} = \frac{A_1+A_2}{H_1+H_2} = \frac{(2a+b)(a+2b)}{9ab}$.

DIBY-65

Find the sum $2017 + \frac{1}{4}\left(2016 + \frac{1}{4}\left(2015 + \cdots + \frac{1}{4}\left(2 + \frac{1}{4}(1)\right)\ldots\right)\right)$

[Ans. : $S = \frac{4}{3}(2017) - \frac{4}{9}\left(1 - \frac{1}{4^{2017}}\right)$]

DIBY-66

Find the sum $1 + 2\left(1 + \frac{1}{50}\right) + 3\left(1 + \frac{1}{50}\right)^2 + \cdots 50$ terms .

[Ans. : 2500]

DIBY-67

Find the sum $\frac{1^2}{2} + \frac{3^2}{2^2} + \frac{5^2}{2^3} + \frac{7^2}{2^4} + \cdots \infty$

[Ans. : 17]

DIBY-68

Let α and β be the roots of equation $px^2 + qx + r = 0, p \neq 0$. If p, q, r, are in A.P. and $\frac{1}{\alpha} + \frac{1}{\beta} = 4$, then the value of $|\alpha - \beta|$ is: [JEE Main-2014]

(A) $\frac{\sqrt{61}}{9}$ (B) $\frac{2\sqrt{17}}{9}$ (C) $\frac{\sqrt{34}}{9}$ (D) $\frac{2\sqrt{13}}{9}$

[Ans. : (D)]

DIBY-69

If b, a, c are in Arithmetic Progression (A.P.) and c, b, a are in Geometric Progression (G.P.), then the sequence a, c, b is in.

(A) Arithmetic Progression (A.P.)
(B) Geometric Progression (G.P.)
(C) Harmonic Progression (H.P.)
(D) Arithmetico-Geometric Progression (A.G.P.)

[Ans. : (C)]

DIBY-70

Find the sum to n terms and also S_∞ for $1+\frac{4}{5}+\frac{7}{5^2}+\frac{10}{5^3}+\cdots$

(A) $S_n=\frac{35}{16}-\frac{12n+7}{16.5^{n-1}}$ and $S_\infty=\frac{35}{16}$

(B) $S_n=\frac{35}{16}-\frac{12n-7}{16\cdot5^{n-1}}$ and $S_\infty=\frac{35}{16}$

(C) $S_n=\frac{25}{16}-\frac{12n+5}{16.5^{n-1}}$ and $S_\infty=\frac{25}{16}$

(D) None of these

[Ans. : (A)]

PROPERTIES OF SIGMA

DIBY-71

The value of $1^2 + 3^2 + 5^2 + \cdots + 25^2$ is : **[JEE Mains-2013]**

(A) 1728 **(B) 1456** **(C) 2925** **(D) 1469**

[Ans. : (C)]

DIBY-72

The sum of the series $1 + 2 \times 3 + 3 \times 5 + 4 \times 7 + \cdots$ upto 11th term is :

[JEE Mains-2019]

(A) 916 **(B) 945** **(C) 915** **(D) 946**

[Ans. : (D)]

DIBY-73

Find the sum of the first n terms of the series $3^2 + 7^2 + 11^2 + \cdots$ up to n terms

(A) $\frac{n}{3}(16n^2 + 12n - 1)$ **(B)** $\frac{n}{3}(16n^2 - 12n + 1)$

(C) $\frac{n}{3}(16n^2 + 12n + 1)$ **(D)** $\frac{n}{2}(16n^2 + 12n - 1)$

[Ans. : (A)]

PROPERTIES OF SIGMA

DIBY-74

Compute the sum of the series whose n^{th} term is given by $T_n = n(n+1)(3n-1)$.

(A) $\frac{n(n+1)(3n^2+3n-2)}{12}$

(B) $\frac{n(n+1)(9n^2+15n+2)}{12}$

(C) $\frac{n(n+1)(n+2)(9n-1)}{12}$

(D) $\frac{n(n+1)(3n^2+n+2)}{4}$

[Ans. : (C)]

DIBY-75

Compute the sum $(31)^2 + (32)^2 + (33)^2 + \cdots \ldots + (50)^2$.

(A) 32,925 **(B)** 33,470 **(C)** 34,185 **(D)** 35,310

[Ans. : (B)]

DIBY-76

The value(S) of the positive integer n for which the quadratic equation, $\sum_{k=1}^{n} (x+k-1)(x+k) = 10n$ has solutions α and $\alpha+1$ for some α is.

(A) 10 **(B)** 11 **(C)** 12 **(D)** 13

[Ans. : (B)]

PROPERTIES OF SIGMA

DIBY-77

The sum $\sum_{k=1}^{20}(1+2+3+\ldots\ldots+k)$ is

[JEE Mains-2020]

[Ans. : 1540]

DIBY-78

Let S_n be the sum to n-terms of an arithmetic progression 3, 7, 11, ... If $40 < \left(\frac{6}{n(n+1)}\sum_{k=1}^{n} S_k\right) < 42$, then n equals

[JEE Mains-2024]

[Ans. : 9]

DIBY-79

The sum of the series $1+2\times3+3\times5+4\times7+\cdots$ upto 11th term is:

[JEE Mins-2019]

(A) 915 (B) 946 (C) 945 (D) 916

[Ans. : (B)]

PROPERTIES OF SIGMA

DIBY-80

Let $S_K = \frac{1+2+\cdots+K}{K}$ and $\sum_{j=1}^{n} S_j^2 = \frac{n}{A}\left(Bn^2 + Cn + D\right)$ where $A, B, C, D \in \mathbb{N}$ and A has least value. Then [JEE Mains-2023]

(A) $A + C + D$ is not divisible by B

(B) $A + B = 5(D - C)$

(C) $A + B + D$ is divisible by 5

(D) $A + B$ is divisible by D

[Ans. : (D)]

DIBY-81

If $\frac{1^3+2^3+3^3+\cdots \text{ upto } n \text{ terms}}{1\cdot3+2\cdot5+3\cdot7+\cdots \text{ upto } n \text{ terms}} = \frac{9}{5}$, then the value of n is [JEE Mains-2023]

[Ans. : 5]

DIBY-82

If $\gcd(m, n) = 1$ and $1^2 - 2^2 + 3^2 - 4^2 + \cdots + (2021)^2 - (2022)^2 + (2023)^2 = 1012m^2n$, then $m^2 - n^2$ is equal to [JEE Mains-2023]

(A) 200 (B) 240 (C) 220 (D) 180

[Ans. : (B)]

PROPERTIES OF SIGMA

DIBY-83

If the sum of the first 15 terms of the series $\left(\frac{3}{4}\right)^3 + \left(1\frac{1}{2}\right)^3 + \left(2\frac{1}{4}\right)^3 + 3^3 + \left(3\frac{3}{4}\right)^3 + \cdots$ is equal to $225k$, then k is equal to: **[JEE Mains-2019]**

(A) 108 (B) 27 (C) 54 (D) 9

[Ans. : (C)]

WRITING T_n FOR SERIES WITH COMMON DIFFERENCE IN AP OR GP

DIBY-84

S_n = 2 + 5 + 14 + 41 + 122 + ... Upto n-terms.

[Ans. : $\frac{3^{n+1}+2n-3}{4}$]

DIBY-85

The sum $1 + 3 + 11 + 25 + 45 + 71 + \cdots$ upto 20 terms, is equal to

[JEE Mains-2025]

(A) 7240 (B) 7130 (C) 6982 (D) 8124

[Ans. : (A)]

DIBY-86

If $S_n = 4 + 11 + 21 + 34 + 50 + \cdots$ to n terms, then $\frac{1}{60}(S_{29} - S_9)$ [JEE Mains-2023]

(A) 226 (B) 220 (C) 223 (D) 227

[Ans. : (C)]

WRITING T_n FOR SERIES WITH COMMON DIFFERENCE IN AP OR GP

DIBY-87

Find the sum of the series 6 + 13 + 22 + 33 + ... up to n terms.

(A) $\frac{n}{6}(2n^2 + 15n + 19)$

(B) $\frac{n}{6}(2n^2 + 13n + 21)$

(C) $\frac{n}{3}(n^2 + 7n + 10)$

(D) $\frac{n}{6}(2n^2 + 11n + 17)$

[Ans. : (A)]

TELESCOPIC METHOD OF DIFFERENCE

DIBY-88

$\frac{1}{3^2-1}+\frac{1}{5^2-1}+\frac{1}{7^2-1}+\cdots+\frac{1}{(201)^2-1}$ is equal to [JEE Mains-2021]

(A) $\frac{101}{404}$ (B) $\frac{25}{101}$ (C) $\frac{101}{408}$ (D) $\frac{99}{400}$

[Ans. : (B)]

DIBY-89

If the sum of series $\frac{1}{1\cdot(1+d)}+\frac{1}{(1+d)(1+2d)}+\cdots+\frac{1}{(1+9d)(1+10d)}$ is equal to 5, then $50d$ is equal to: [JEE Mains-2024]

(A) 20 (B) 5 (C) 15 (D) 10

[Ans. : (B)]

DIBY-90

The sum $\sum_{n=1}^{21}\frac{3}{(4n-1)(4n+3)}$ is equal to [JEE Mains-2022]

(A) $\frac{7}{87}$ (B) $\frac{7}{29}$ (C) $\frac{14}{87}$ (D) $\frac{21}{29}$

[Ans. : (B)]

TELESCOPIC METHOD OF DIFFERENCE

DIBY-91

The sum of 10 terms of the series $\frac{3}{1^2\times2^2}+\frac{5}{2^2\times3^2}+\frac{7}{3^2\times4^2}+\cdots$ is **[JEE Mains-2021]**

(A) $\frac{143}{144}$ (B) 1 (C) $\frac{99}{100}$ (D) $\frac{120}{121}$

[Ans. : (D)]

DIBY-92

If $\frac{1}{2\times3\times4}+\frac{1}{3\times4\times5}+\frac{1}{4\times5\times6}+\cdots+\frac{1}{100\times101\times102}=\frac{k}{101}$, then $34k$ is equal to

[JEE Mains-2022]

[Ans. : 286]

DIBY-93

If $\sum_{k=1}^{10}\frac{k}{k^4+k^2+1}=\frac{m}{n}$, where m and n are co-prime, then $m+n$ is equal to

[JEE Mains-2022]

[Ans. : 166]

TELESCOPIC METHOD OF DIFFERENCE

DIBY-94

The sum to 10 terms of the series $\frac{1}{1+1^2+1^4}+\frac{2}{1+2^2+2^4}+\frac{3}{1+3^2+3^4}+\cdots$ is

[JEE Mains-2023]

(A) $\frac{59}{111}$ (B) $\frac{55}{111}$ (C) $\frac{56}{111}$ (D) $\frac{58}{111}$

[Ans. : (B)]

DIBY-95

If $a_n=\frac{-2}{4n^2-16n+15}$, then $a_1+a_2+\cdots+a_{25}$ is equal to: [JEE Mains-2023]

(A) $\frac{51}{144}$ (B) $\frac{49}{138}$ (C) $\frac{50}{141}$ (D) $\frac{52}{147}$

[Ans. : (C)]

DIBY-96

The sum of the series $\frac{1}{1-3\cdot1^2+1^4}+\frac{2}{1-3\cdot2^2+2^4}+\frac{3}{1-3\cdot3^2+3^4}+\ldots$ up to 10 terms is

[JEE Mains-2024]

(A) $\frac{45}{109}$ (B) $-\frac{45}{109}$ (C) $\frac{55}{109}$ (D) $-\frac{55}{109}$

[Ans. : (D)]

TELESCOPIC METHOD OF DIFFERENCE

DIBY-97

If the sum of the first 20 terms of the series $\frac{4.1}{4+3.1^2+1^4}+\frac{4.2}{4+3.2^2+2^4}+\frac{4.3}{4+3.3^2+3^4}+\frac{4.4}{4+3.4^2+4^4}+\cdots$ is $\frac{m}{n}$, where m and n are coprime, then $m+n$ is equal to

(A) 423 (B) 420 (C) 421 (D) 422

[Ans. : (C)]

DIBY-98

The sum $\sum_{k=1}^{n}\frac{1}{(k+1)\sqrt{k}+k\sqrt{k+1}}$ is equal to

[Ans. : $\frac{\sqrt{n+1}-1}{\sqrt{n+1}}$]

TELESCOPIC METHOD OF DIFFERENCE

DIBY-99

Find the sum $\sum_{n=1}^{\infty} \frac{6^n}{(3^n-2^n)(3^{n+1}-2^{n+1})}$ [Ans. (2)]

DIBY-100

Find the sum of the series $\frac{1}{3^2+1}+\frac{1}{4^2+2}+\frac{1}{5^2+3}+\frac{1}{6^2+4}+\cdots\infty$. [Ans. (13/36)]

DIBY-101

Find the sum $\sum_{n=2}^{\infty} \frac{3n^2+1}{(n^2-1)^3}$ [Ans. (09/16)]

DIBY-102

Let $S=\frac{\sqrt{1}}{1+\sqrt{1}+\sqrt{2}}+\frac{\sqrt{2}}{1+\sqrt{2}+\sqrt{3}}+\frac{\sqrt{3}}{1+\sqrt{3}+\sqrt{4}}+\cdots+\frac{\sqrt{n}}{1+\sqrt{n}+\sqrt{n+1}}=10$.

Then find the value of n. [Ans. (24)]

INEQUALITY

DIBY-103

Minimum value of sum of real numbers $a^{-5}, a^{-4}, 3a^{-3}, 1, a^8$ & a^{10} (where $a > 0$)

[IIT-JEE-2011]

[Ans. 8]

DIBY-104

If $a > 0, b > 0, c > 0$, then show that : $(a+b+c)\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right) \geq 9$ [JEE-1984]

DIBY-105

For positive real number a, b & c show that $\left((1+a)(1+b)(1+c)\right)^7 > 7^7 a^4 b^4 c^4$ [JEE-2004]

DIBY-106

Let $x, y > 0$. If $x^3y^2 = 2^{15}$, then the least value of $3x + 2y$ is [JEE Mains-2022]

(A) 30 (B) 32 (C) 36 (D) 40

[Ans. (D)]

INEQUALITY

DIBY-107

The minimum value of $f(x) = a^{a^x} + a^{1-a^x}$,, where $a, x \in R$ and $a > 0$, is equal to

[JEE Mains-2021]

(A) $a + \frac{1}{a}$ (B) $2a$ (C) $a + 1$ (D) $2\sqrt{a}$

[Ans. D]

DIBY-108

Let a, b, c and d be positive real numbers such that $a + b + c + d = 11$. If the maximum value of $a^5 b^3 c^2 d$ is 3750β, then the value of β is

[JEE Mains-2023]

(A) 110 (B) 108 (C) 90 (D) 55

[Ans. C]

DIBY-109

If the minimum value of $f(x) = \frac{5x^2}{2} + \frac{\alpha}{x^5}, x > 0$ is 14, then the value of α is equal to

[JEE Mains-2022]

(A) 32 (B) 64 (C) 128 (D) 256

[Ans. (C)]

EXPONENTIAL & LOG SERIES

DIBY-110

Simplify : $S = \frac{2}{1!} + \frac{4}{3!} + \frac{6}{5!} + \frac{8}{7!} + \ldots\ldots\infty$

[Ans. (e)]

DIBY-111

$$S = 1 + \frac{1+2}{2!} + \frac{1+2+2^2}{3!} + \frac{1+2+2^2+2^3}{4!} + \ldots\ldots\infty$$

[Ans. $(e^2 - e)$]

DIBY-112

The sum $1 + \frac{1+3}{2!} + \frac{1+3+5}{3!} + \frac{1+3+5+7}{4!} + \ldots\ldots$ upto ∞ terms, is equal to

[JEE Mains-2025]

(A) 6e (B) 4e (C) 3e (D) 2e

[Ans. (D)]

EXPONENTIAL & LOG SERIES

DIBY-113

The sum of the series $\sum_{n=1}^{\infty} \frac{n^2+6n+10}{(2n+1)!}$ is equal to: [JEE Mains-2021]

(A) $\frac{41}{8}e + \frac{19}{8}e^{-1} - 10$

(B) $\frac{41}{8}e - \frac{19}{8}e^{-1} - 10$

(C) $\frac{41}{8}e + \frac{19}{8}e^{-1} + 10$

(D) $-\frac{41}{8}e + \frac{19}{8}e^{-1} - 10$

[Ans. : B]

DIBY-114

If $0 < x < 1$ and $y = \frac{1}{2}x^2 + \frac{2}{3}x^3 + \frac{3}{4}x^4 + \cdots$, then the value of e^{1+y} at $x = \frac{1}{2}$ is: [JEE Mains-2021]

(A) $2e^2$

(B) $\frac{1}{2}e^2$

(C) $2e$

(D) $\frac{1}{2}\sqrt{e}$

[Ans. : B]

It is never about end result

It's always about Journey

THANK You

#futureIITians